FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IX

JEE (Advanced)-2019

PAPER – 1

TEST DATE: 28-04-2019

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-D.**

1. ***Section-A (01– 06, 19 – 24, 37 - 42)*** *contains 18 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen, both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*

   *Negative Marks : **–2** In all other cases.*

   Section-A (07 – 10, 25 – 28, 43 - 46) *contains 12 questions. Based on this section contains* ***TWO (02)*** *paragraphs. Based on each paragraph, there are* ***TWO (02)*** *questions.*
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-D (11 – 18, 29 – 36, 47 – 54)*** *contains 24 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

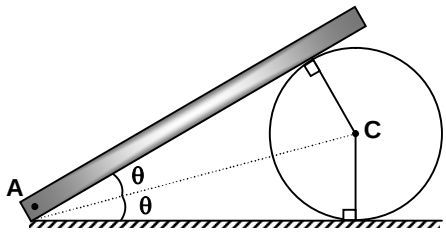
PART – I

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

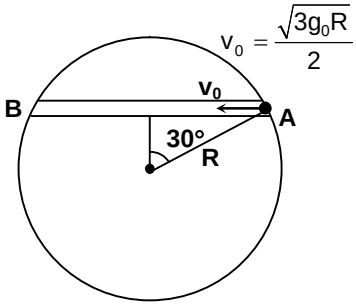
1. A particle starts moving in a plane with constant tangential acceleration $a_t = 2$ m/s^2 and normal acceleration $a_n = t^4$. Then choose the correct option(s).
   (A) The acceleration of the particle is $2\sqrt{5}$ m/s^2 after having covered a distance S = 2m.
   (B) The acceleration of the particle is $2\sqrt{2}$ m/s^2 after having covered a distance S = 2m.
   (C) The radius of curvature of its trajectory is 4m after having covered a distance S = 1m.
   (D) The radius of curvature of its trajectory is 2m after having covered a distance S = 1m.

2. A solid sphere of mass m = 2.4 kg and radius R = 0.5 m is placed on a rough horizontal surface. A uniform rod AB of mass $m_0 = 2.5$ kg and length $\ell = 1$m hinged at one end 'A' is resting on the sphere as shown. The rod makes an angle '2θ' = 74° with the horizontal surface initially and then the system is released from rest. There is no slipping between the sphere and horizontal surface and there is no friction between the rod and sphere. Then choose the correct option(s).



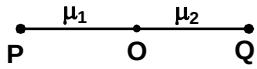
   (A) The acceleration of sphere just after release is (approx) 1 m/s^2.
   (B) The acceleration of sphere just after release is (approx) 2 m/s^2.
   (C) The normal force between the rod and sphere just after release is (approx) 6.5 N.
   (D) The normal force between the rod and sphere just after release is (approx) 3.5 N.

3. A tunnel is dug across the earth along a chord AB as shown. A particle is projected inside the tunnel from end 'A' with an initial velocity $v_0 = \frac{\sqrt{3g_0R}}{2}$, there is no friction between the tunnel and the particle. Then choose the correct option(s). (where g_0 = gravitational acceleration on the surface of Earth, R = radius of earth)



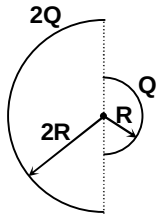
   (A) The time taken by the particle to move from end 'A' to other end 'B' of the tunnel is $\frac{\pi}{3}\sqrt{\frac{R}{g_0}}$.
   (B) The time taken by the particle to move from end 'A' to other end 'B' of the tunnel is $\frac{2\pi}{3}\sqrt{\frac{R}{g_0}}$.
   (C) The maximum speed of the particle inside the tunnel is $\sqrt{g_0R}$.
   (D) The maximum speed of the particle inside the tunnel is $\sqrt{2g_0R}$.

4. Two strings of equal radii but mass per unit length μ_1 = 0.9 kg/m and μ_2 = 0.1 kg/m are joined together and stretched between two rigid supports as shown. A sinusoidal wave pulse of amplitude 3 cm is incident from end 'P' and transmission and reflection both take place at the joint 'O'. There is no loss of power during propagation of the wave. Then choose the correct option(s).



(A) The amplitude of the reflected wave is 1.5 cm.
(B) The amplitude of the transmitted wave is 4.5 cm
(C) The percentage of incident power transmitted to the second string is 60%.
(D) The percentage of the incident power reflected to the first string is 25%.

5. Two hemispherical thin shells made of insulating materials are concentrically arranged in a free space as shown. The radii of the smaller and larger hemispheres are 'R' and '2R' and they carry positive charges 'Q' and '2Q' respectively. The charges are uniformly distributed over the surfaces of the shells. Then choose the correct option(s).



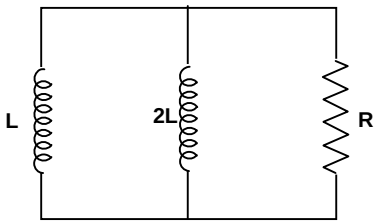
(A) The electrostatic force between the hemispherical shells is $\frac{Q^2}{8\pi\varepsilon_0 R^2}$

(B) The electrostatic force between the hemispherical shells is $\frac{Q^2}{16\pi\varepsilon_0 R^2}$

(C) The net electric potential at the centre 'O' of the shells is $\frac{Q}{2\pi\varepsilon_0 R}$

(D) The net electric potential at the centre 'O' of the shells is $\frac{3Q}{2\pi\varepsilon_0 R}$

6. A parallel circuit consists of the two ideal inductances 'L' and '2L' and a resistance 'R' as shown. At t = 0, the same current 'I_0' is flowing in both the inductances in the same direction. Then choose the correct option(s).



(A) The energy stored in inductance 'L' in steady state is $\frac{LI_0^2}{18}$.

(B) The energy stored inductance '2L' in steady state is zero.

(C) The net heat dissipated in the resistor is $\frac{3LI_0^2}{2}$

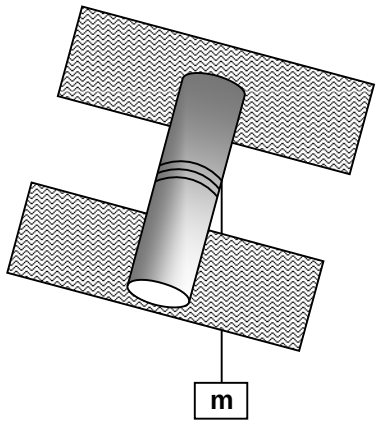
(D) The net heat dissipated in the resistor is $\frac{4}{3}LI_0^2$

(Paragraph Type)

This section contains **TWO** paragraphs. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 7 and 8

Charge 'Q' is uniformly distributed on the non conducting surface of a thin uniform long hollow cylinder of mass 'm' radius 'R' and length 'ℓ'. It is placed on two stationary rough horizontal planks. A string is wound over the cylinder whose other end is attached to a block of same mass 'm'. When the system is released from rest, the cylinder starts rolling without slipping relative to the planks. Now answer the following questions. (Take $\frac{\mu_0 Q^2}{4\pi\ell} = m$)



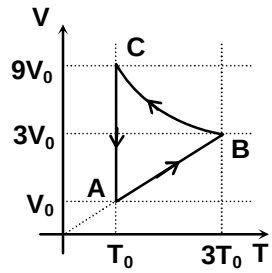
7. The acceleration of the hollow cylinder is
   (A) 2g
   (B) $\frac{3g}{4}$
   (C) $\frac{g}{2}$
   (D) $\frac{g}{4}$

8. The tension in the string is
   (A) $\frac{3mg}{4}$
   (B) $\frac{3mg}{2}$
   (C) $\frac{mg}{4}$
   (D) $\frac{mg}{2}$

Paragraph for Question Nos. 9 and 10

Two moles of an ideal mono-atomic gas is taken through a cyclic process ABCA as shown in the V-T diagram. In the process BC, VT = constant. The temperature at A is T_0 = 300 K. [take ℓn(3) = 1.1]

Then answer the following questions.



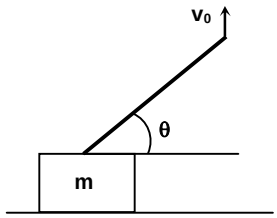
9. The work done by the gas during the process 'BC' is
   (A) 600 R
   (B) 900 R
   (C) 1200 R
   (D) 1500 R

10. The efficiency (in percentage) of the cycle is
   (A) 12
   (B) 18
   (C) 24
   (D) 36

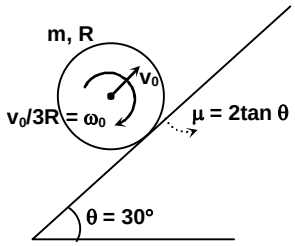
SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

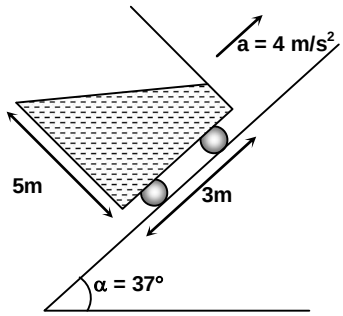
11. A small block of mass m = 2 kg is placed on a smooth horizontal surface. A massless inextensible string of length ℓ = 1.5 m with one end attached to the block is initially held in horizontal position. With what constant velocity v_0 (in m/s) the free end of the string is moved in vertical upward direction so that the block will leave the horizontal surface, when the string makes an angle $\theta = 37°$ with the horizontal surface. (take g = 10 m/s^2 and tan 37° = 0.75)



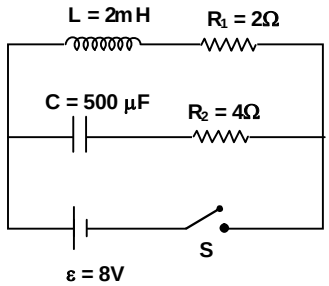
12. A solid sphere of mass m = 3 kg and radius R = 10 cm is projected up an inclined plane of inclination $\theta = 30°$ with an initial velocity v_0 = 15 m/s and initial angular velocity $\omega_0 = \frac{v_0}{3R}$ as shown. The coefficient of friction between the sphere and inclined plane is $\mu = 2\tan\theta$. Then find the velocity (in m/s) of the sphere, when it will start rolling without slipping relative to the inclined plane.



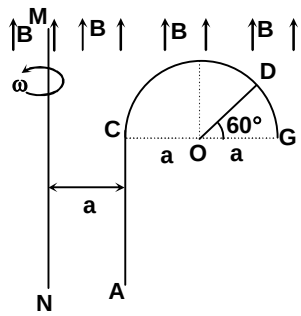
13. An open rectangular tank of base area (3m × 4m) and height 5 m filled with water upto height 4m is accelerated up an inclined plane of inclination $\alpha = 37°$ with an acceleration a = 4 m/s^2. Find the volume (in m^3) of water that will spill over the tank. (take g = 10 m/s^2)



14. In the adjacent circuit the switch 'S' is closed at t = 0. Then find the current (in amp) through the battery at t = 1 × 10^{-3} sec. (take e^{-1} = 0.37, e$^{-0.5}$ = 0.60)

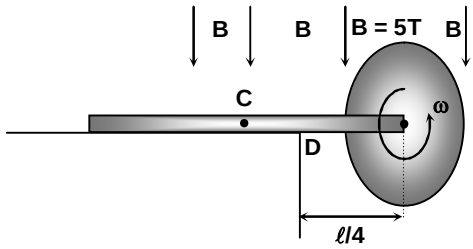


15. A conductor ACDG consists of a straight segment AC = 0.5 m and a semicircular arc CDG of radius a = 20 cm. A uniform vertical magnetic field B = 5 Tesla is existing in the region. Now the conductor is rotated with a constant angular velocity ω = 10 rad/s about a vertical axis MN parallel to the magnetic field as shown. Then find the emf (in volt) induced between the points A and D of the conductor as shown.



16. The pitch of a screw gauge is 1 mm and its cap is divided into 100 divisions. When nothing is placed between its studs, the zero of circular scale is 4 divisions above the reference line. When a wire is placed between its studs, the main scale reading is 2 mm and 46th division of circular scale coincides with the reference line. The length of wire is 4.76 cm. Find the curved surface area (in cm^2) of the wire in true significant figures.

17. Charge q = 0.5 coulomb is uniformly distributed over the surface of a non conducting circular disc of mass m = 0.2 kg and radius R = 0.5 m. The disc is mounted at one end of a uniform rod of mass m_0 = 1 kg and length ℓ = 2m so that the plane of disc is perpendicular to the length of rod and the rod is placed on a rough horizontal surface with ℓ/4 length of the rod is protruded over the edge of the surface as shown. A uniform vertical downward magnetic field B = 5T is existing in the region. Find the minimum counter clockwise (when viewed from right side of the disc) angular velocity 'ω' (in rad/s) with which the disc is rotated about its horizontal axis so that the rod will topple about the edge (at point D) of the horizontal surface.



18. When an α-particle with kinetic energy K = 11 MeV collides with a stationary lithium nucleus, the following nuclear reaction takes place

$^4He + {}^7Li \rightarrow {}^{10}B + n$

If Q – value of this reaction is E = -2.86 Mev. Find the kinetic energy (in MeV) of the neutron outgoing at right angle to the incoming direction of α-particle.

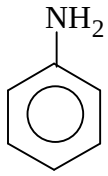
Chemistry PART – II

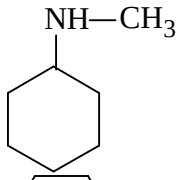
SECTION – A
(One or More than one correct type)

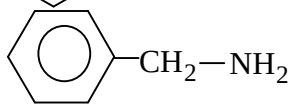
This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

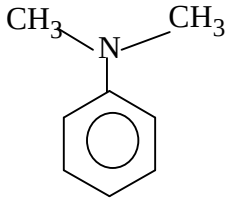
19. For a real gas obeying van der waal's equation which of the following are true?
(A) Internal energy of gas is dependent on temperature only
(B) If Z > 1 then forces of repulsion are dominant
(C) At high pressure $Z = \left(1 + \frac{Pb}{RT}\right)$
(D) Work done in expansion of a real gas is more than for an ideal gas under isothermal conditions.

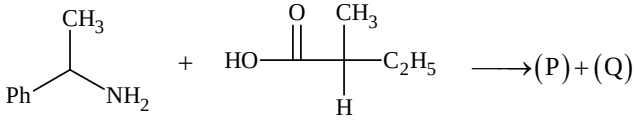
20. Which of the following compounds give carbylamine when reacted with $CHCl_3$ and ethanolic KOH ?

(A) 

(B) 

(C) 

(D) 

21. 

$(R-\text{form})$ $(R+S, \text{racemic mixture})$

The correct statement regarding compound (P) & (Q) is/are:
(A) Both are optically active in nature
(B) (P) & (Q) are diastereomers
(C) (P) & (Q) are meso compounds
(D) Out of (P) & (Q) one is optically active and other is optically inactive

22. Which of the following order is/are incorrect ?
(A) $H_3PO_4 > H_3PO_3 > H_3PO_2$ (reducing character)
(B) $N_2O < NO < N_2O_3 < N_2O_5$ (oxidation state of nitrogen atom)
(C) $NH_3 > PH_3 > AsH_3 > SbH_3$ (Basic)
(D) $SbH_3 > NH_3 > AsH_3 > PH_3$ (reducing character)

23. Following two equilibria are simultaneously established in a container

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g) \; ; \; CO(g) + Cl_2(g) \rightleftharpoons COCl_2(g)$$

If some Ni(s) is introduced in the container establishing another equilibrium

$$4CO(g) + Ni(s) \rightleftharpoons Ni(CO)_4(g)$$

Which among the following statements is/are definitely true about this system?
(A) $COCl_2$ concentration decreases as the new equilibrium is established.
(B) As Ni is a solid it will not have any impact on the existing two equilibrium reactions.
(C) PCl_3 concentration decreases to re-attain the equilibrium.
(D) PCl_5 concentration increases to attain the new equilibrium.

24. Which of the following molecules is/are planar?
(A) $F_2C = C = CF_2$
(B) $F_2B - CH = CH - BF_2$
(C) $(SiH_3)_3N$
(D) $H_2N - NH_3$

(Paragraph Type)

This section contains **TWO** paragraphs. Based on each paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

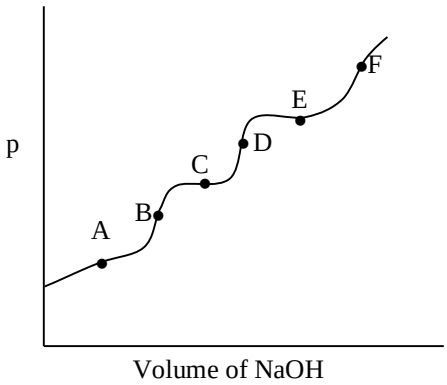
Paragraph for Question Nos. 25 and 26

When H_3PO_4 solution is titrated against NaOH reactions occurs in three steps

$H_3PO_4 + NaOH \rightarrow NaH_2PO_4 + H_2O$
$NaH_2PO_4 + NaOH \rightarrow Na_2HPO_4 + H_2O$
$Na_2HPO_4 + NaOH \rightarrow Na_3PO_4 + H_2O$

If 0.1M of 300ml H_3PO_4 is titrated using 0.1M NaOH then titration curve pH vs volume of NaOH is given below

$Ka_1 (H_3PO_4) = 10^{-3}$ $Ka_2(H_3PO_4) = 10^{-8}$
$Ka_3(H_3PO_4) = 10^{-13}$



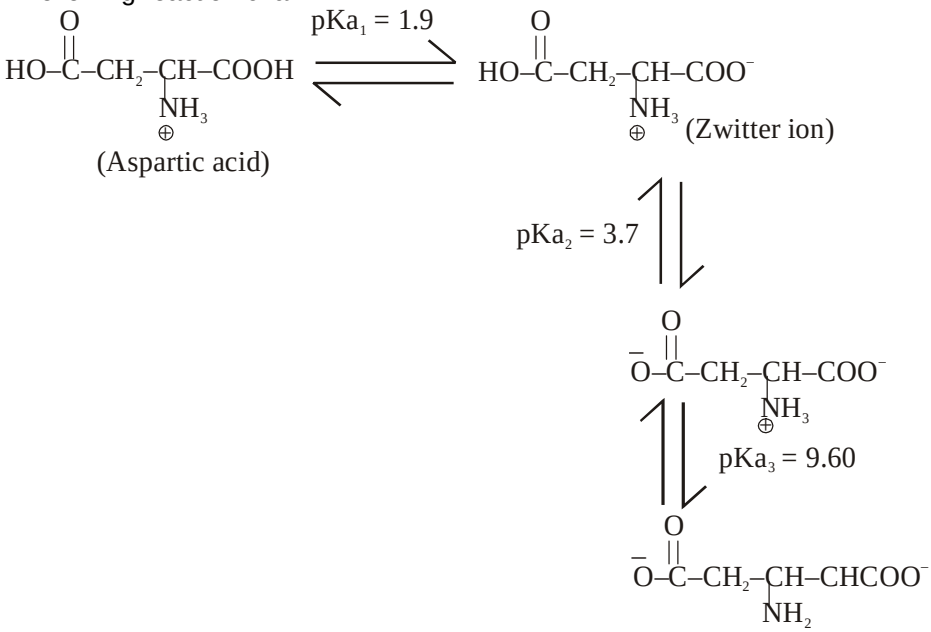
25. What will be the pH of the solution when 50ml of NaOH is added to 0.1M, 300ml H_3PO_4 solution ($M_{NaOH} = 0.1$)?
(A) 3
(B) 2.3
(C) 3.36
(D) 4.6

26. What is the composition of solution at point C?
(A) Buffer of $NaH_2PO_4 + Na_2HPO_4$
(B) Buffer of $NaH_2PO_4 + H_3PO_4$
(C) Only NaH_2PO_4
(D) Only Na_2HPO_4.

Paragraph for Question Nos. 27 and 28

The glycine as well as other amino acids, is amphoteric, means it contains an acidic functional group and a basic functional group. The $-\overset{\oplus}{N}H_3$ is an acidic functional group and basic functional group is $-COO^-$. In a strongly acidic medium the species present is $-\overset{\oplus}{N}H_3CH_2COOH$ (R) as pH is raised, a proton is removed from this species. Amino acids are crystalline solids it does not melts but on being heated it decomposes. These are practically soluble in water but insoluble in non polar organic solvents. These properties are attributed to fact that the stable form of amino acids is a Zwitter ion or inner salt.

27. In following reaction chain



The isoelectric point of aspartic acid is
(A) 2.8
(B) 4.8
(C) 5.75
(D) 1.9

28. When glycine is dissolved in a solution having pH = 9; at 25°C; it will move towards — in electrolytic cell combination
(A) Anode
(B) Cathode
(C) Will not move towards any electrode
(D) None of above

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

29. When fluorine reacts with 2-methyl propane in presence of light gives 14% tert-butyl fluoride and 86% isobutylfluoride. What will be the ratio of relative reactivity of a tertiary hydrogen to primary.

30. If a water tank of 10000 Ltr. contains 55.5 g of $CaCl_2$, 47.5 g of $MgCl_2$, 84 g of $MgSO_4$ and 81.6 g of $CaSO_4$, then what will be the degree of hardness for 1 ltr. of water from this tank? (Assuming density of solution in tank = 1 g/ml)

31. Biacetyl, the flavouring that makes margarine taste "just like butter," is extremely stable at room temperature, but at 200°C it undergoes a first order break down with a half-life of 12.0 min. An industrial flavour enhancing process requires that a biacetyl flavoured food be heated briefly at 200°C. How long (in minutes) can the food be heated and retain 85% of its butter flavour? (log 1.176 = 0.0706)

32. An element crystallises in fcc lattice. If the effective number of atoms of the element per unit cell in the crystal is decreased by 0.25% due to Schottky defect then the density of crystal is (At. Wt. = 50 gm and edge length = 0.50 nm)

33. A thin circular-disk earrings 5.0 cm in diameter is plated with a coating of gold 0.20 mm thick from an Au^{3+} bath. How many days (approx.) does it take to deposit the gold on one side of one earring if the current is 0.02 A (MW. of Au = 197 g, density of Au = 19.3 g/cm, 1 F = 96500 C)?

34. 3129 kJ of heat is liberated when 60 grams of ethane is completely burnt in oxygen. ΔH^0_f for CO_2 and $H_2O(\ell)$ are $-395kJ$ and $-286kJ$ respectively. The magnitude of heat of formation of ethane is

35. 0.5 g mixture of $K_2Cr_2O_7$ and $KMnO_4$ was treated with excess of KI in acidic medium. Iodine so liberated required 100 ml of 0.15 N solution of sodium thiosulphate for titration. The percentage of $K_2Cr_2O_7$ in the mixture is (MW of $K_2Cr_2O_7$ = 294 and $KMnO_4$ = 158)

36. At 800°C, the following equilibrium is established as:

$F_2(g) \rightleftharpoons 2F(g)$

The composition of the equilibrium may be determined by measuring the rate of effusion of the mixture through a pinhole. It is found that at 800°C and 1 atm the mixture effuses 1.6 times as fast as SO_2 effuses under similar conditions (At. wt. of F = 19). The value of K_P for the equilibrium is

Mathematics PART – III

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. In a plane rectangular co-ordinate system xRy there are three points $A\left(0, \frac{4}{3}\right)$, B(–1, 0) and C(1, 0). The distance from point P to line BC is the geometric mean of the distances from this point to lines AB and AC. Then the locus of point P is
(A) $2x^2 + 2y^2 + 3y - 2 = 0$
(B) $2x^2 - 2y^2 - 3y + 2 = 0$
(C) $8x^2 - 17y^2 + 12y - 8 = 0$
(D) $8x^2 + 17y^2 - 12y - 8 = 0$

38. Let S = {1, 2, 3, 4, ..., 2070} and A be the number of subsets of S whose sum of elements in S is divisible by 9. If $A = \frac{2^a\left(2^b + 1\right)}{c}$, then
(A) c = 9
(B) b = 1840
(C) a = 233
(D) b = 1837

39. If the lines $a\bar{z} + \bar{a}z + b = 0$ and $c\bar{z} + \bar{c}z + d = 0$ are mutually perpendicular, where a and c are non-zero complex numbers and b and d are real numbers, then
(A) $a\bar{a} + c\bar{c} = 0$
(B) $a\bar{c}$ is purely imaginary
(C) $\text{Arg}\left(\frac{a}{c}\right) = \pm\frac{\pi}{2}$
(D) $\frac{a}{\bar{a}} = \frac{c}{\bar{c}}$

40. Of the following statement which of the below are correct, when n is natural number
(A) probability ${}^nC_7$ is divisible by $7 = \frac{1}{7}$
(B) probability ${}^nC_7$ is divisible by $12 = \frac{91}{144}$
(C) probability ${}^nC_7$ is divisible by $7 = \frac{13}{49}$
(D) probability ${}^nC_7$ is divisible by $12 = \frac{77}{144}$

41. Let θ be an acute angle such that the equation $x^2 + 4x\cos\theta + \cot\theta = 0$ involving variable x has multiple roots. Then the measure of θ in radians is/are
(A) $\frac{\pi}{6}$
(B) $\frac{\pi}{12}$

(C) $\frac{5\pi}{12}$

(D) $\frac{7\pi}{12}$

42. $I_{(m,n)} = \int_0^{\pi} \cos mx \cdot \sin nx \, dx$ where, m, n are integers, then which of the following is/are correct?

(A) $I_{(16,2)} = 0$

(B) $I_{(8,5)} = -\frac{10}{39}$

(C) $I_{(5,2)} = -\frac{4}{21}$

(D) $I_{(3,7)} = 0$

(Paragraph Type)

This section contains **TWO** paragraphs. Based on each paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 43 and 44

Read the following write up carefully and answer the following questions:

Let π_1 denotes the equation of the plane to which the vector (1, 1, 0) is normal and which contains the line L whose equation is $\vec{r} = \hat{i} + \hat{j} + \hat{k} + \lambda\left(\hat{i} - \hat{j} - \hat{k}\right)$ and π_2 denotes the equation of the plane containing the line L and a point whose position vector (0, 1, 0)

43. The vector along a normal to the plane π_2 is

(A) $\hat{i} + 2\hat{j} - \hat{k}$

(B) $\hat{i} - 2\hat{j} + \hat{k}$

(C) $\hat{i} + 2\hat{j} + \hat{k}$

(D) $\hat{i} + \hat{j} - 2\hat{k}$

44. The acute angle between π_1 and π_2 is

(A) $\tan^{-1}1 - \tan^{-1}\left(\sqrt{2} - 1\right)$

(B) $\tan^{-1}1 - \tan^{-1}\left(2 - \sqrt{3}\right)$

(C) $\tan^{-1}1 - \tan^{-1}\left(\sqrt{2} + 1\right)$

(D) $\tan^{-1}1 - \tan^{-1}\left(2 + \sqrt{3}\right)$

Paragraph for Question Nos. 45 and 46

Read the following write up carefully and answer the following questions:

Let z_1 and z_2 are two complex numbers such that $|z_1| = a$ and $|z_2| = b$ satisfying the equation $3z_1^2 - 2z_1z_2 + 2z_1^2 = 0$ and also $\text{Re}\left(\frac{z_1 - 2}{z_1 + 2}\right) = 0$. Also P, Q and O are points in complex plane corresponding to z_1, z_2 and origin respectively

45. Eccentricity of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, is

(A) $\frac{1}{\sqrt{2}}$

(B) $\frac{1}{\sqrt{3}}$

(C) $\frac{1}{2}$

(D) $\frac{1}{3}$

46. Area of triangle OPQ is

(A) $\sqrt{5}$

(B) $2\sqrt{5}$

(C) $3\sqrt{5}$

(D) $4\sqrt{5}$

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

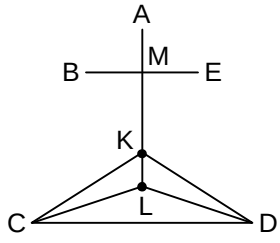
47. The value of $\lim\limits_{n\to\infty} n^2 \int\limits_{-\frac{1}{n}}^{\frac{1}{n}} (2019\sin x + 2021\cos x)|x|dx$ is equal to

48. Given $a^2 + b^2 + c^2 \le 16$ and $S = 3a + 4b + 10c$ where $a, b, c \in R$. If the maximum value of S is $\sqrt{N}$, then the value of N is equal to

49. There is a point $A(x_1, y_1)$ on the curve $f(x) = x^2$ and a point $B(x_2, y_2)$ on the curve $g(x) = -\frac{8}{x}$ where $x_1 > 0$ and $x_2 > 0$. If the line AB is tangent to both these curve at these points respectively, then the value of $(x_1 + y_1 + x_2)$

50. If all the roots of the equation $x^{1008} + a_1x^{1007} + a_2x^{1006} + ... + a_{1006}x^2 - 1008x + 1 = 0$ are real, then $\frac{a_2}{|a_1|}$ is

51. The diagram shows a maze used for testing and training mice. A mouse begins at A and at each junction as represented by a dot has equal chances of choosing any of the new path available. The mouse can move in any direction but cannot travel along any path more than once, the probability of the mouse reaching point, C after passing through more than 3 junctions is $\frac{K}{L}$ (where K and L are relatively coprime. Then the value of L – K



52. $\prod_{k=1}^{6} \sin\frac{k\pi}{7} = \frac{\ell}{m}$ (where ℓ, m are relatively coprime), then $\ell + m$ is equal to

53. If $y^2 = x^2 - x + 1$ and $I_n = \int \frac{x^n}{y} dx$ & $AI_3 + BI_2 + CI_1 = x^2y$ for all $x \in R$. Then the value of A + B + C is

54. For any real number be denote by f(b) is the maximal value of $\left|\sin x + \frac{2}{3+\sin x} + b\right|$. The minimal value of f(b) is $\frac{p}{q}$, where p and q are relatively co-prime positive integers. Then the value of p + q